

Lecture 9: Principles of Stereoscopic Display

- Close one eye and try to judge the distance to an object
- Then open both eyes and try again

Ask:

“Which felt more accurate?”

Explain:

This happens because **our brain uses two-eye vision to calculate depth** — this is the foundation of stereoscopic displays.

Human Depth Perception

Explain simply:

Depth perception is the brain’s ability to judge how far objects are.

Two types of depth cues:

1. Monocular Cues (one eye enough)

- Size of object
- Overlap (occlusion)
- Perspective
- Light & shadow
- Motion parallax

Example:

A car far away looks smaller than a nearby car.

2. Binocular Cues (require both eyes)

This is most important for VR:

- Binocular disparity
- Convergence

Tell students:

Stereoscopic displays simulate binocular vision artificially.

Binocular Disparity

Definition:

Binocular disparity is the difference between the images seen by the left and right eye.

Explain with simple idea:

- Eyes are about **6–7 cm apart**
- Each eye sees a slightly different image
- Brain combines both → depth is perceived

Simple class demo:

Ask students to:

- Hold a finger in front of their face
- Close left eye → observe position
- Close right eye → observe position shift

Explain:

That shift is **disparity**. VR headsets create two slightly different images to trick the brain.

Link to VR:

- VR headset shows:
 - One image for left eye
 - One image for right eye
 - This creates 3D effect
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Convergence

Definition:

Convergence is the inward rotation of eyes when focusing on nearby objects.

Explain:

- When object is close → eyes rotate inward
 - When object is far → eyes become parallel
 - VR system must:
 - Adjust object depth
 - Match convergence with disparity
 - If mismatch → eye strain, discomfort, VR sickness
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How This Works in Stereoscopic Displays (VR Headsets)

Explain practical connection:

In VR headset:

- Two screens (or split screen)
- Each eye gets different perspective
- Lenses help focus
- Brain fuses both → depth illusion

Human Vision	VR System
Two eyes	Two rendered views
Binocular disparity	Slight image shift per eye
Convergence	Simulated via object depth
Brain fuses images	Perceived 3D scene

Explain problem:

If VR system poorly handles disparity/convergence → nausea, headache, fatigue.

This links beautifully to:

- Why rendering accuracy matters
 - Why stereoscopic principles are essential in AR/VR design
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Recap + Quick Questions

Recap:

- Depth perception comes from monocular + binocular cues
- Binocular disparity = difference between left/right eye images
- Convergence = eye rotation to focus on depth
- VR headsets simulate both to create 3D illusion

Quick oral questions:

- Why does closing one eye reduce depth accuracy?
- What creates binocular disparity in VR?
- Why does poor convergence cause discomfort?